

Optimization Techniques in Power System (BT EE610)



हरियाणा केंद्रीय विश्वविद्यालय
CENTRAL UNIVERSITY OF HARYANA
NAAC ACCREDITED 'A' GRADE UNIVERSITY

Unit-I

History of Optimization Techniques

By
Dr. Manish Kumar
Asth. Professor
Department of Electrical Engineering
SOE&T, CUH

History of Optimization Techniques

History of finding optimal designs

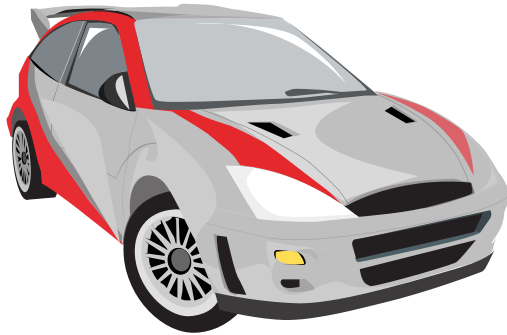


Design experimentally

A car with efficient aerodynamics

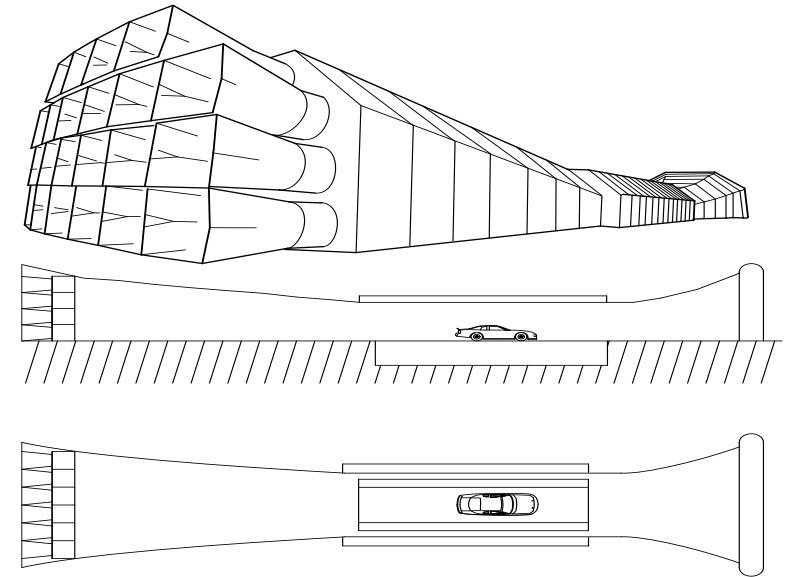


Bob (Engineer)



Bob wants to design a car with minimum drag

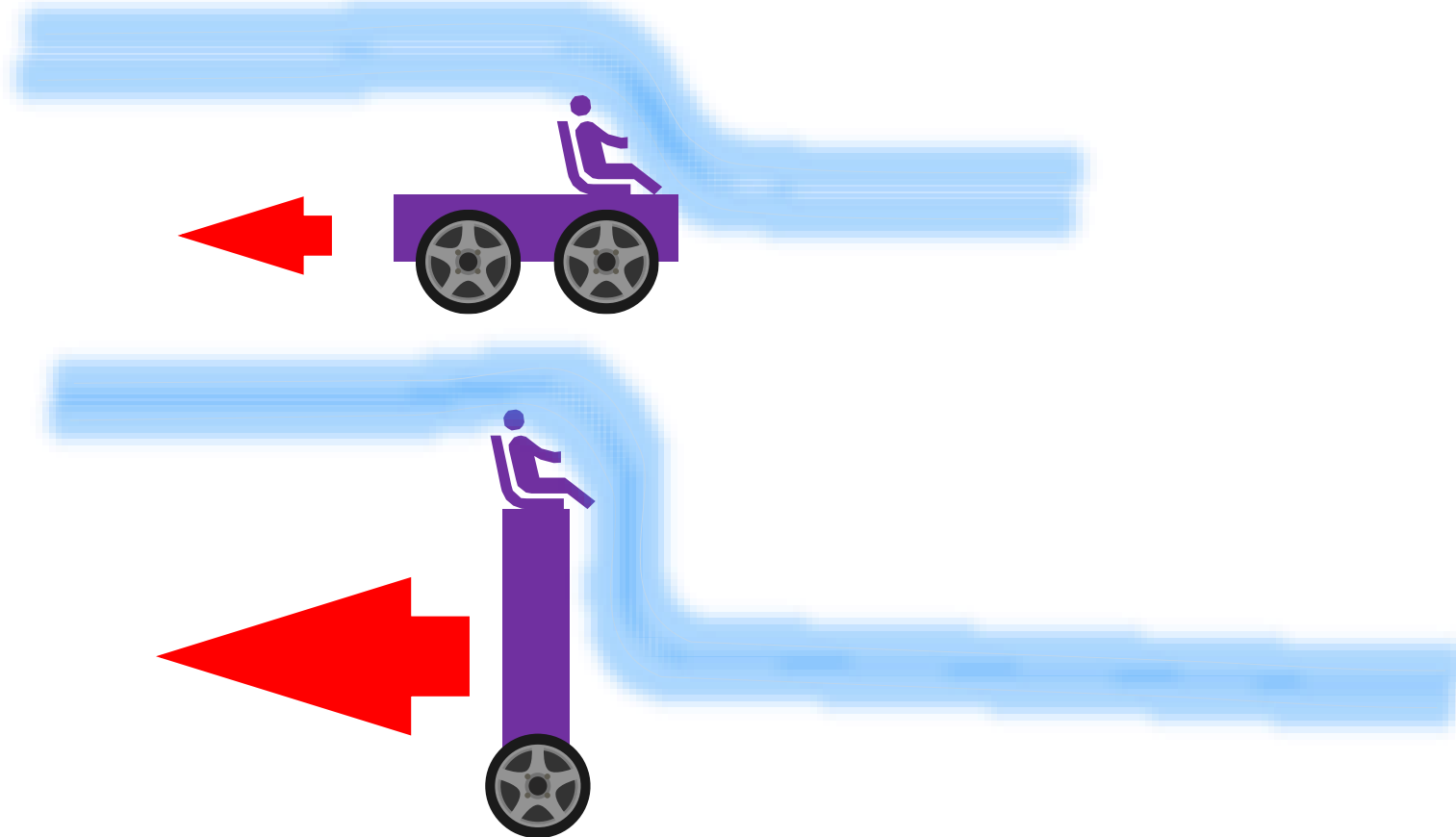
Wind tunnel to analyse aerodynamic forces (drag)

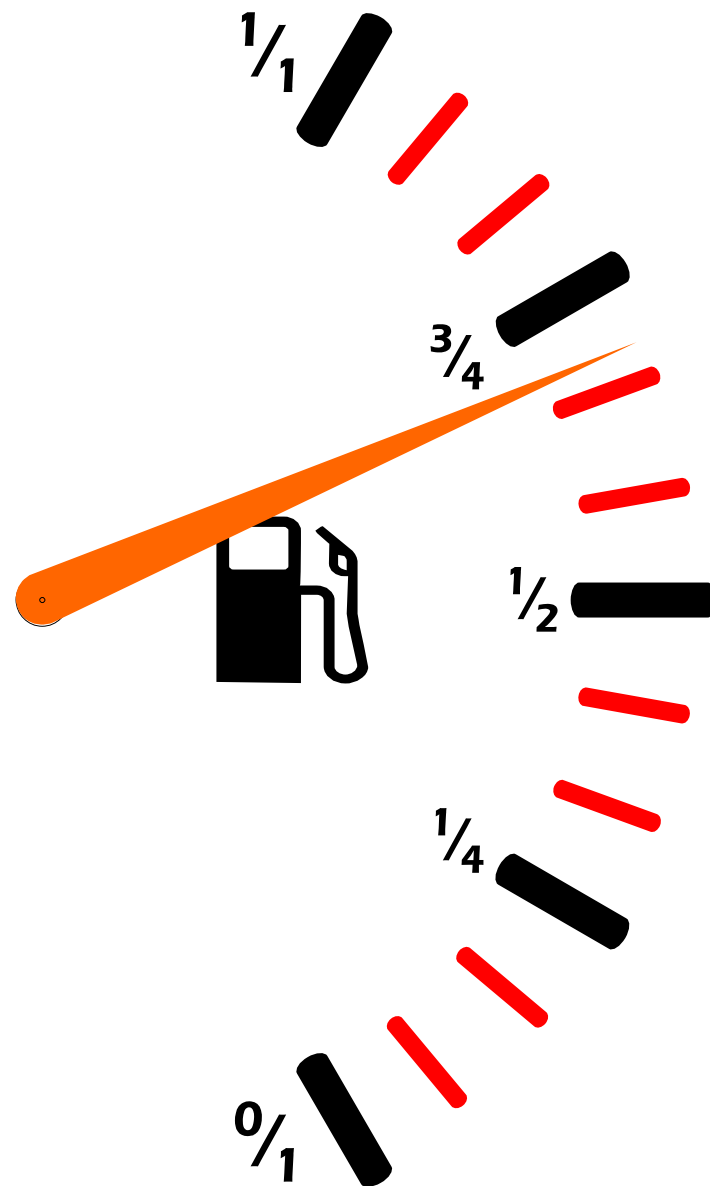
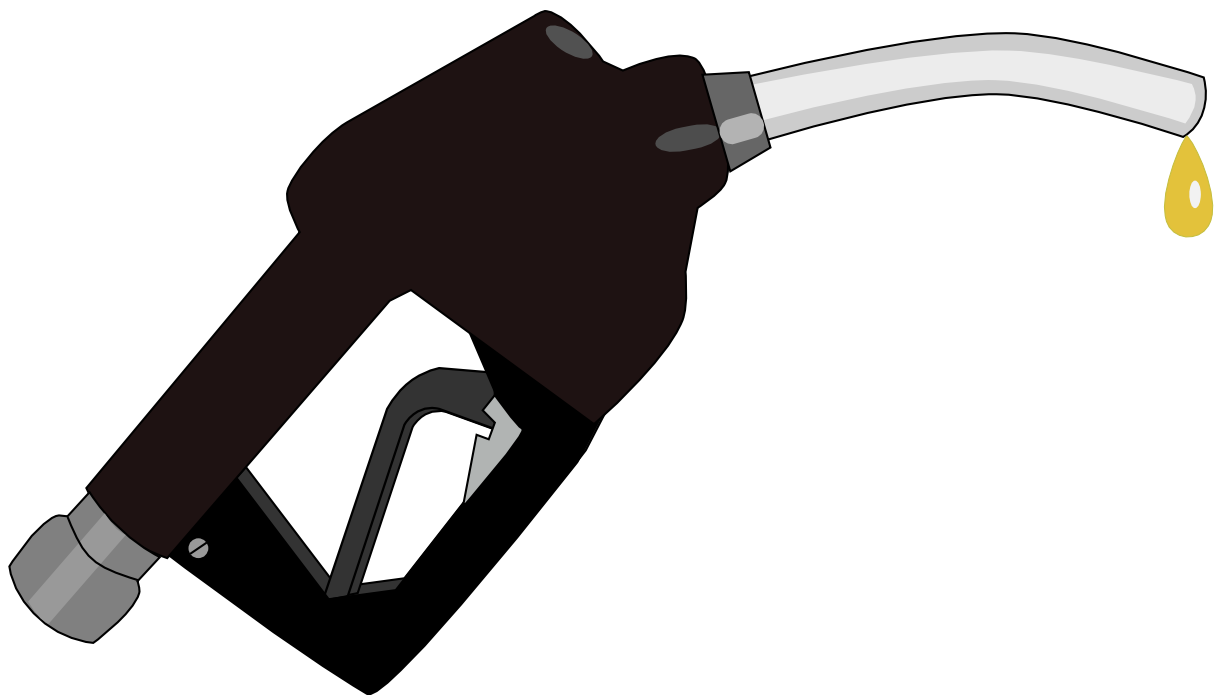


He needs to build prototypes and a wind tunnel to calculate drag

What is drag?

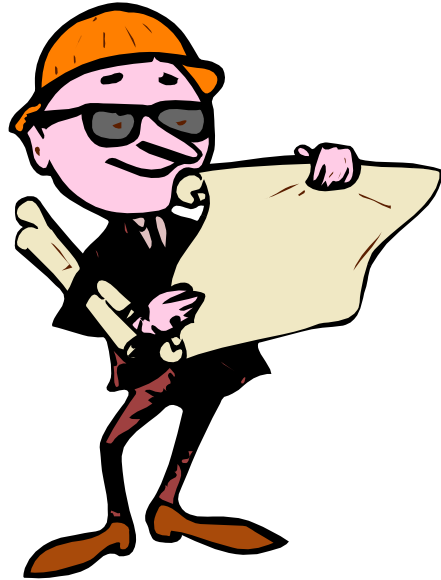
- Drag (air resistance) is an aerodynamic force acting opposite to the relative motion of any object.





Design experimentally

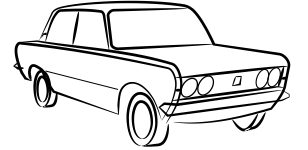
Coefficient of drag (C_d)



$$C_d = 0.85$$



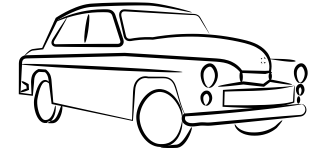
$$C_d = 0.90$$



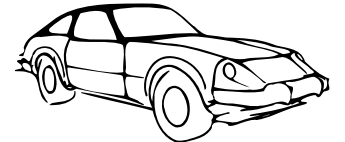
$$C_d = 0.50$$



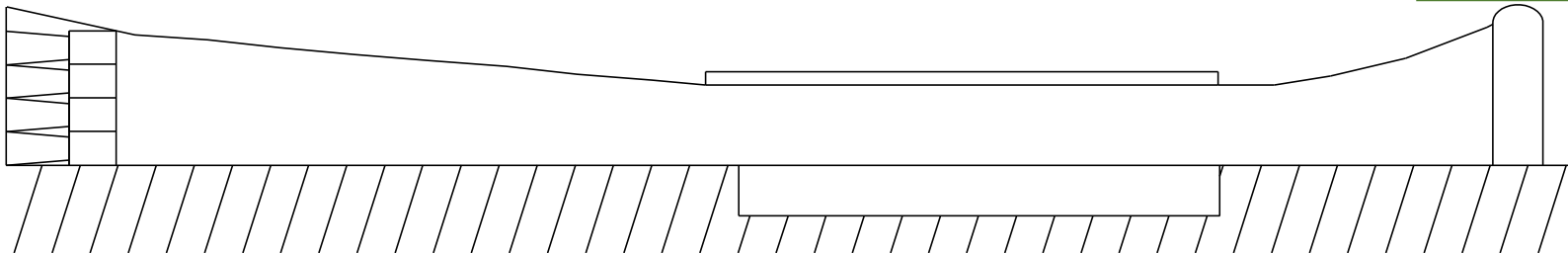
$$C_d = 0.45$$



$$C_d = 0.40$$

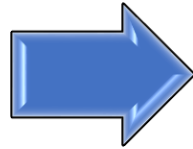


$$C_d = 0.32$$

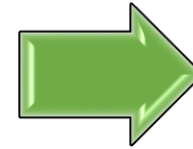


Design experimentally

- Tedious
- Expensive
- Time consuming
- Human involvement
- Not accurate

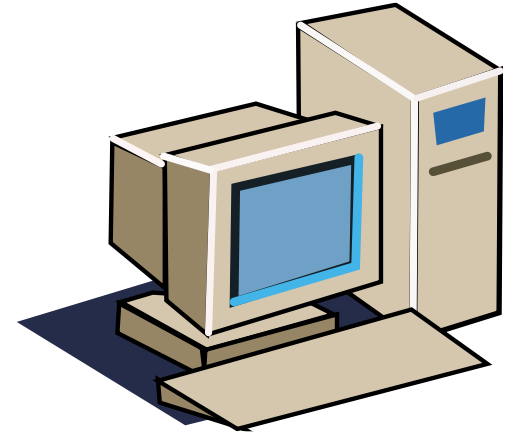
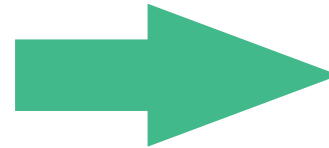
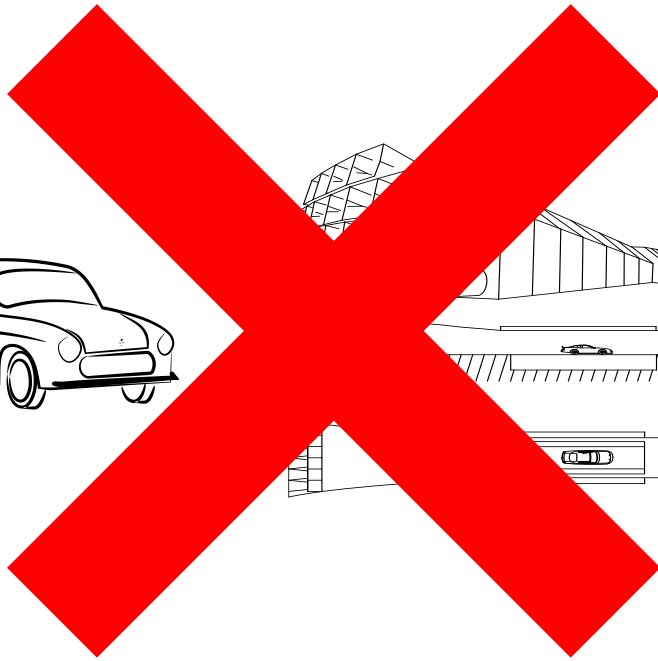


Design with simulated models

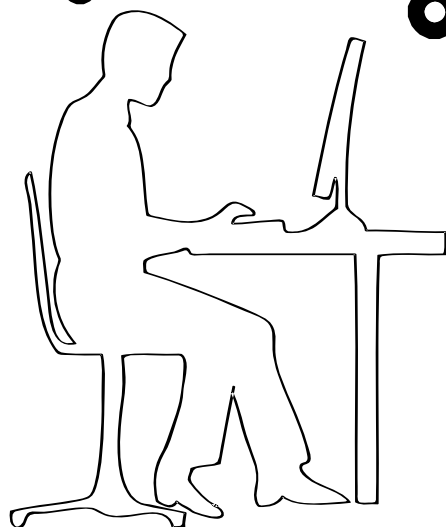
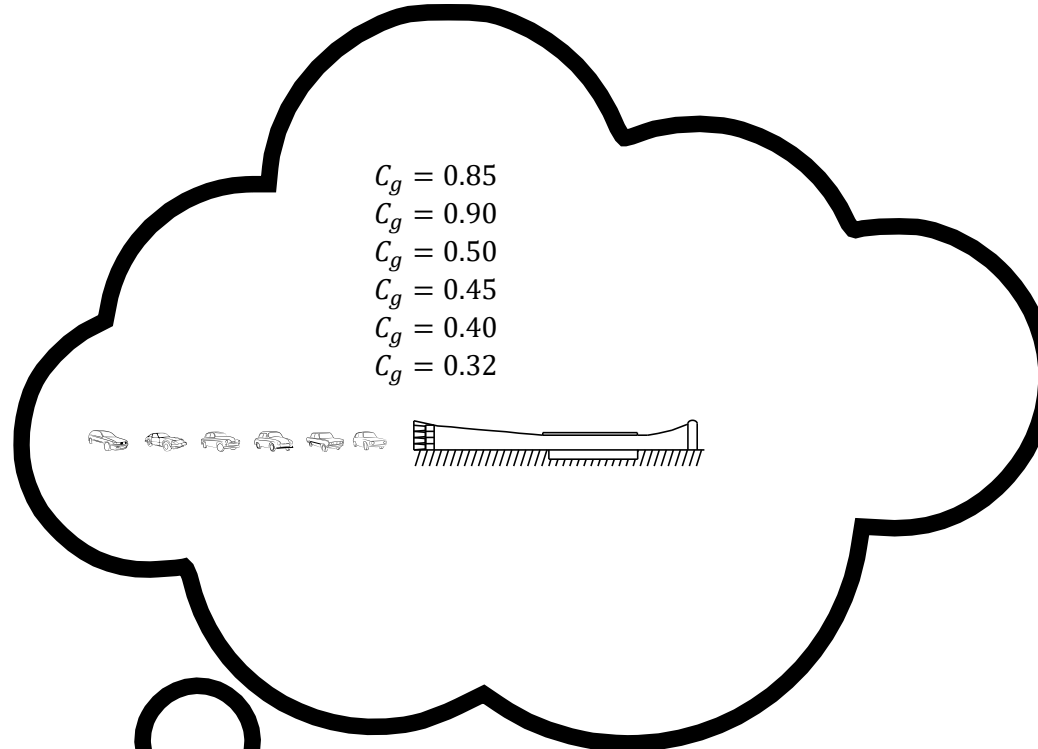
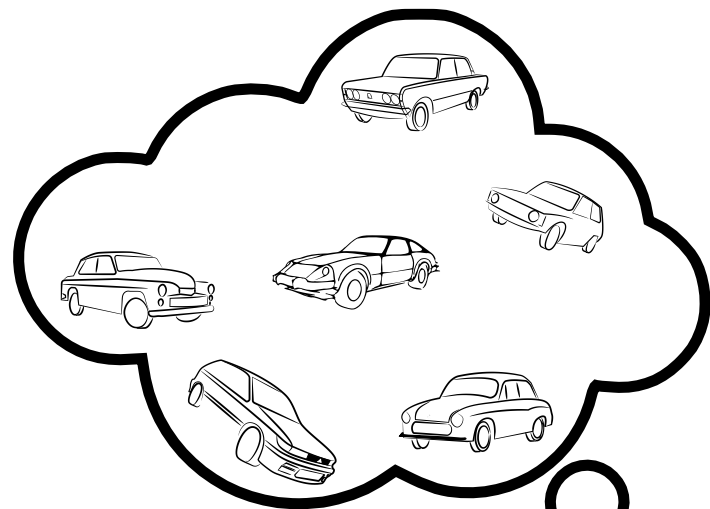


Design with optimization algorithms

Design with simulated
models



Design with simulated models

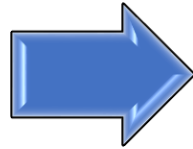






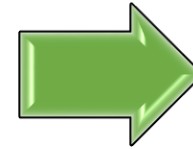
Design experimentally

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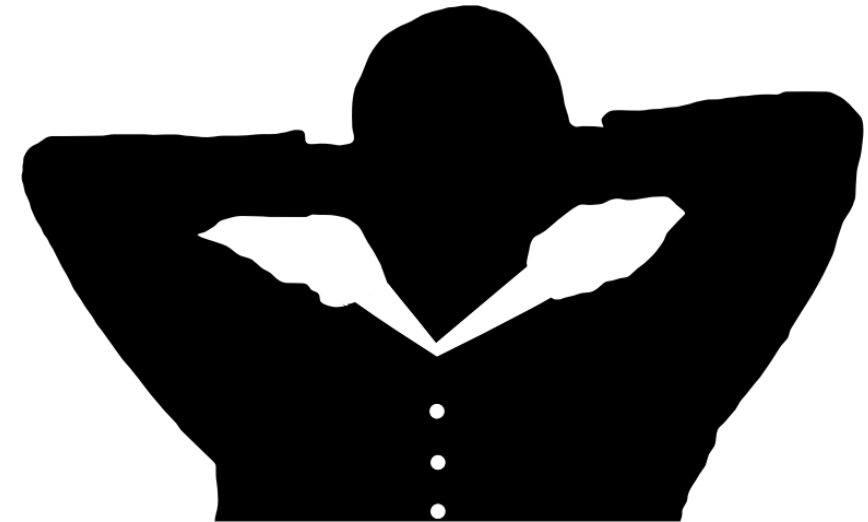
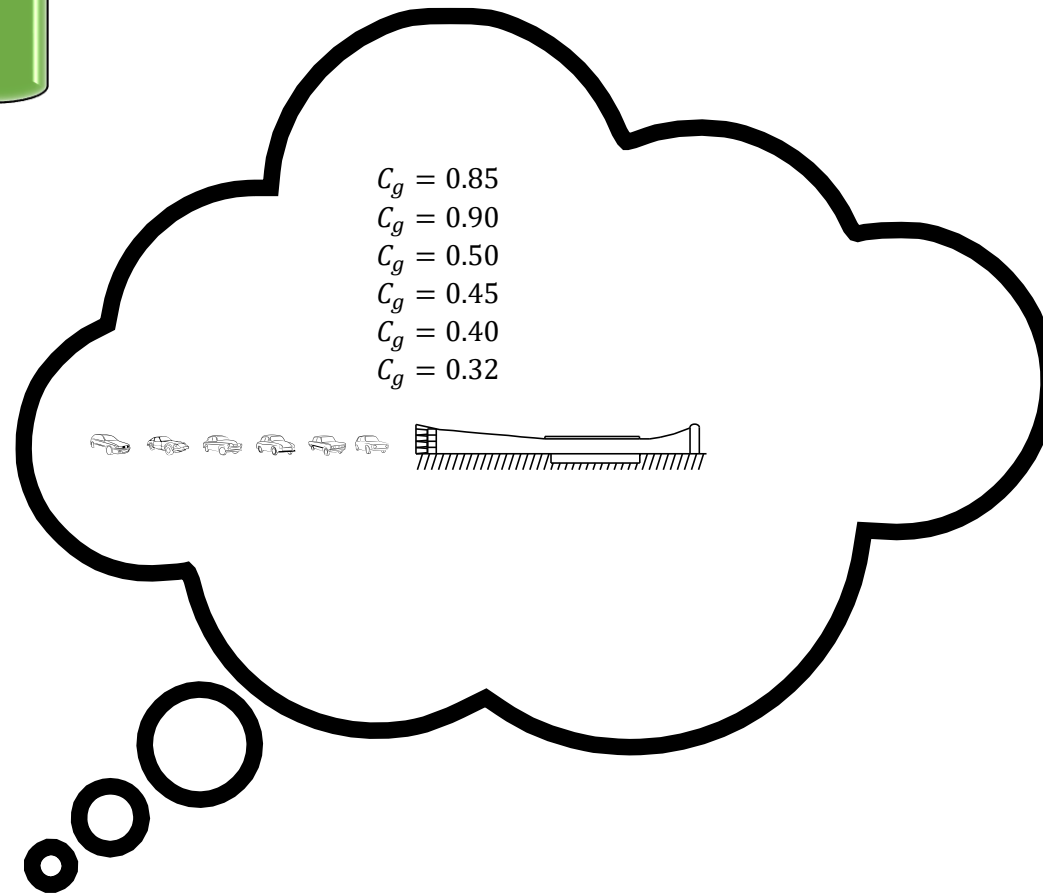
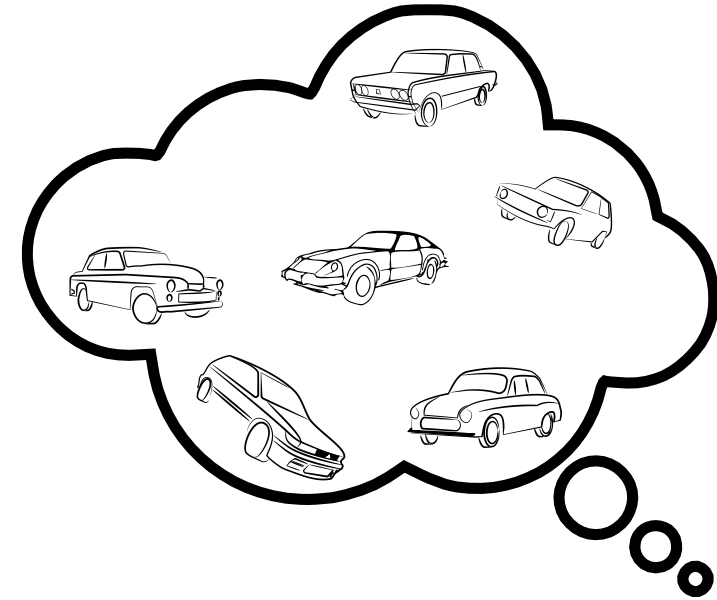
Design with simulated models

- Fast and cheap
- Slow design
- Medium human involvement
- Error prone



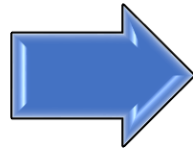
Design with optimization algorithms

Design with optimization algorithms



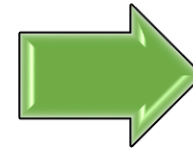
Design experimentally

- Tedious
- Expensive
- Time consuming
- Human involvement
- Not accurate



Design with simulated models

- Fast modeling
- Slow design process
- Medium human involvement
- Error prone



Design with optimization algorithms

- Fast modeling
- Fast design process
- Automated (minimum human involvement)
- Low error
- Complex optimization algorithm
- Difficulties of solving real world problems

